MODULE 0
No-Bullshit Entrepreneurship

VENTURE LABS

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UNTERNEHMER TUM

Course Outline

	Module	Outcome
Block I	M01 Team & Foundations · M02 Finding Ideas · M03 Problem Space · M04 Validating Problems	Team formed → one tested idea → pain map → landing page with committed threshold
Block II	M05 Solution Space · M06 Prototyping · M07 The Kill Decision · M08 First Users	48h build → 10 stranger reactions → Go/Pivot/Kill → first outside-€
Block III	M09 Revenue · M10 Product & Growth · M11 Founder Health · M12 Fundraising & Demo Day	Pricing tested → PMF measured → pitch v1 → €1,000 verified + continuation decision

How we conduct each sessions

Session	Topics	Type
0:00–0:20	Check-In & Recap	Group Discussion & Input Session
0:20–0:50	Progress Presentations	Group Work & Presentations
0:50–2:00	New Content	Input Session
2:00–3:20	Workshop	Group Work & Self-Paced Learning
3:20–3:30	Check-Out	Group Discussion

Recommended	
Home Assignment	Group Assignment

Optional	
Case Study	Self-Paced or Group Work
Readings	Self-Paced

Some Housekeeping

1 Together we grow: Coachability

We give candid and productive feedback

We listen and seek to understand

2 We act as we say: Accountability

We reliably stick to our words if its in our hands

We try and learn from failures if its not

3 Give first, dream first: Bias-to-Action

We prefer building and experimenting over understanding every aspect

We know that you get more if you give

Today

Why this course exists — and what it demands (10')

IDEA 1 · Startups are systems — they die of self-inflicted wounds (15')

IDEA 2 · Evidence beats opinions (10')

IDEA 3 · Speed is your only advantage (10')

IDEA 4 · The contract: real users, real revenue, kill criteria (10')

Case · Juicero: the $120M machine for a job your hands do better (30')

Workshop · Startup Autopsy + First Signal (90')

Wrap-up · your first 3 problem hypotheses (5')

IDEA 1: Startups are systems

Cash in, cash out, a clock running in the background — while (cash > 0), you exist.

Most startups don't get murdered by competitors. They die of self-inflicted wounds: building things nobody wants.

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Startups as Systems

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Think of a Startup as a System

Input: an idea, some money, and your time

Process: build something, show it to people, learn, repeat

Output: either a product people pay for -- or nothing

Constraint: you will run out of money before you run out of ideas

Feedback: the market tells you if you're wrong. Listen or die.

"while (cash > 0) { iterate(); }
That's the entire startup algorithm. Everything else is details."

-- First-time founder

while (cash > 0)
{ iterate(); }

That's the entire startup algorithm. Everything else is details.

Cash, Burn Rate, Runway

Cash

How much money is in the bank. Not revenue (money from customers), not valuation (what someone says your company is worth). Cash is the literal euros in your account.

Burn Rate

How much you spend per month. Salaries, cloud hosting, rent, tools — everything. If you spend €50k/month, your burn rate is €50k.

Runway

Cash ÷ Burn Rate = months until you're dead.

Example: €200k in the bank, €50k/month burn = 4 months of runway. When runway hits zero, the company dies.

Why this matters

Runway is the only number that kills you. You can survive bad marketing, a weak product, even a mediocre team — temporarily. But zero cash = game over. Every decision you make as a founder either extends or shortens your runway.

Revenue, ARR, Churn

Revenue

Money customers pay you. The only signal that isn't noise. Everything else — website traffic, social media followers, press mentions — is a vanity metric until someone pays.

ARR (Annual Recurring Revenue)

Your monthly revenue × 12. If customers pay you €10k/month, your ARR is €120k. This is the number investors (called VCs, or venture capitalists) use to benchmark your progress.

Churn

The % of customers who stop paying each period. 5% monthly churn means you lose half your customers every year. If churn > growth, you're filling a leaky bucket.

Break-Even

When revenue = costs. You stop dying. Most startups never get here — they either grow past it or run out of cash first.

Rule of thumb: if churn is higher than growth, you're filling a leaky bucket. Fix the bucket before pouring in more water.

Product-Market Fit

The holy grail. When your product solves a real problem for people who will pay for it.

Before PMF: nothing else matters. Not your brand, not your process, not your org chart.

After PMF: everything gets easier. Users tell their friends. Revenue grows without heroics.

How you know you don't have it: you have to convince people to use your product.

How you know you do: people get angry when you take it away.

"Product-market fit means being in a good market with a product that can satisfy that market."

-- Marc Andreessen

IDEA 2: Evidence beats opinions

Your idea is a hypothesis. Your mom's praise is not data. Your friends' likes are not data.

This course runs on the Hierarchy of Evidence: opinions < conversations < behavior < money.

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The base rate — read before you fall in love

Most startups fail. Most VALIDATED startups still fail. Outcomes are power-law: a few winners pay for everything.

The cases in this course are mostly winners — because they teach decisions, not because winning is typical.

Survivorship bias is the tax on every startup story you've ever heard. Today you meet the graveyard: Juicero.

Kill criteria aren't pessimism — they're how you lose small and try again fast.



a 20-company VC portfolio, returns drawn to scale — #1 pays for everything

What the evidence says

This course's method is tested: in a randomized controlled trial with 116 startups, founders taught to state and test hypotheses like scientists earned MORE revenue and pivoted more decisively (Camuffo, Cordova, Gambardella & Spina 2020, Management Science; replicated since).

Treat your idea as the hypothesis, the market as the lab, revenue as the read-out.

Everything else — interviews, landing pages, prototypes, pricing tests — is just apparatus.

Nobody cares about your TAM slide.

They care about evidence that customers will pay you. Show traction, not projections.

IDEA 3: Speed is your only advantage

You have no brand, no money, no distribution. What you have: nothing to lose and a one-week decision cycle.

Every week here is one full build–measure–learn loop. Incumbents need a quarter for that.

Speed, concretely

One week = one loop: hypothesis Monday → evidence by Friday → decision in check-in.

Fast beats perfect because iterations compound: 12 honest loops beat 3 polished ones.

The €10k teams weren't smarter. They started asking for money in week 8, not week 12.

Why now — and why here

AI collapsed the cost of building: a funded team's six months in 2020 is a weekend prototype today.

Europe's moment: capital, talent and political will lining up behind technical founders (more in the guest deck).

Munich: TUM Venture Labs, UnternehmerTUM, CDTM, EXIST — Personio, Celonis and Isar Aerospace started inside this ecosystem. You're sitting in it.

IDEA 4: The contract

Real users. Real revenue. Pre-committed kill criteria — written BEFORE every test.

Past cohorts ended with €1,000–10,000 in verified revenue. Not essays. That's the bar you're signing up for.

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How you're graded (published today)

Evidence quality 30% · decision discipline 30% · verified traction 30% · communication 10%.

A rigorously argued KILL with a documented evidence pack can be an A. A zombie project cannot.

Revenue rules: outside your social graph · dashboard/invoice-verified · repeatable beats consulting.

Full rubric + the Revenue Ladder: resources/GRADING_RUBRIC.md — read it tonight.

Words we use precisely (you'll be graded in this vocabulary)

VALIDATION — somebody opens their wallet. Everything before that is SIGNAL.

SIGNAL — interviews, signups, clicks: earns you the right to run the next test. Valuable. Not validation.

PROTOTYPE — a quickly built slice of the product that users are willing to PAY for. Not a mockup, not a demo.

CUSTOMER — a user who pays. A user who doesn't pay is a fan.

KILL — an evidence-based decision to stop. In here, a win — and gradeable as one.

Full glossary: resources/GLOSSARY.md. We will correct your vocabulary in check-ins. Kindly. Every time.

The Simulation Game

Optional Fun Part to get you familiar with Startup Terms

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The Slingshot: An AI Startup Simulation

You run an AI startup for 16 quarters (4 simulated years).

Complete 3 milestones before running out of cash. Milestones = prove tech, find customers, scale.

Compete against 3 AI-controlled rivals racing to the same milestones.

Choose from 8 AI sectors (healthcare, robotics, fintech, etc.) and 12 founder types.

550+ events drawn from real startup research. No right answers, only trade-offs.

Most players fail their first attempt. That's the point.

NOTE: Don‘t take the simulation part too seriously. It‘s a game first and foremost.

How It Works

Each Quarter

Get 6 Attention Points (AP) -- your CEO bandwidth.

Allocate AP across Research, Product, Sales, People.

You can't do everything. That's the constraint.

Bank up to 2 unused AP for next quarter.

Staff growth gives +1 to +3 bonus AP.

The Trade-offs

Publish research openly? +credibility, -competitive advantage.

Hire fast? +capacity, -culture, -cash.

Take VC money? +runway, -equity, -control.

Ship buggy? +speed, +technical debt.

Every decision has a cost. The game makes you feel it.

Key Takeaways

Startups are systems with one hard constraint: finite cash

Every decision is a trade-off. There are no right answers, only informed ones.

The market is the only honest feedback mechanism. Everything else is opinions.

Vocabulary matters: if you can't name it, you can't discuss it, debug it, or fix it.

Failure is data. Your second attempt will be fundamentally different.

Sources

Paul Graham (2012). Startup = Growth. paulgraham.com/growth.html

Marc Andreessen (2007). Product/Market Fit. pmarchive.com

Salter, Baruffaldi & Bignone (2025). The Slingshot: An AI Startup Simulation. CC BY-NC-SA 4.0.

The Slingshot Teaching Guide v3. github.com/AI-Startup-Game

Further Reading

Paul Graham -- all essays at paulgraham.com. Start with 'Do Things That Don't Scale' and 'How to Start a Startup'.

Eric Ries -- The Lean Startup. The MVP/pivot framework everyone references.

Peter Thiel -- Zero to One. Contrarian thinking about competition and monopoly.

Steve Blank -- The Four Steps to the Epiphany. Where 'customer development' comes from.

Brad Feld -- Venture Deals. Read this before you read any term sheet.

Y Combinator Startup School -- free at startupschool.org. Best lectures per minute.

a16z blog -- future.com. VC perspective on markets and tech trends.

Workshop: Startup Autopsy + First Signal (90 min)

Agenda lives in M00_Workshop_Startup-Autopsy.md — one source of truth.

Must produce: completed autopsy sheet + your 3 problem hypotheses + 3 First-Signal messages SENT to real people.

Kill criteria: you can't name a single real human who has the problem.

This week — or you're behind

Read PG's 'How to Get Startup Ideas'. Identify 3 problems YOU personally experience.

Talk to 2 real people about their daily frustrations — names + one verbatim quote each in the Toolkit.

Homework: M00_Homework.md. Next week we form teams around the problems you bring.

The course contract starts now: evidence, not opinions.

Reflection: The Results – Rose, Thorn, Bud

What was the breakthrough today?



What was the hardest part today?



What is an idea/feature you didn't have time to finish, but want to explore later?



Thank You!

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APPENDIX — original slides (21 restored as images)

These pages from the pre-rework deck are not in the current flow (incl. the Slingshot detail block and funding stack). Source: Instructor/original-decks/M00_Lecture_ORIGINAL.pdf.

Mark each: KEEP (design pass rebuilds it editable) or KILL. Nothing is deleted without your call. Note: appendix images show the pre-fix text (Zurich/Tübingen etc.).

Course Outline

Session	Module	Outcome
Zurich	Module 1: Teambuilding & Entrepreneurial Basics	Form a team and find problems you love
Zurich	Module 2: Finding Ideas	Data Room: Know the constrains in which you can excel
Zurich	Module 3: The Problem Space	Presentation: Understand the dynamics of your problem
Zurich	Module 4: Validating Problem Spaces	Landingpage: Validate your problem space with real users
Tübingen	Module 5: The Solution Space	Click-Dummy: Build and validate solutions with real users
Tübingen	Module 6: Rapid Prototyping	Prototype: Ship a functional prototype to test users
Tübingen	Module 7: Rapid Validation	MVP: Receive feedback on your solution from potential customers
Tübingen	Module 8: Finding First Users	Live-Solution: Converting your first occasional users into customers
Munich	Module 9: From Prototype to Product	Live-Product: Grow your product to support 1.000 MAU
Munich	Module 10: Product Management	Create a growth roadmap & investor presentation
Munich	Module 11: Founder Mental Health & Well-Being	Reflect on your needs and energy
Munich	Module 12: Presenting & Fund Raising	1on1 Mentoring for next Steps

Check-In & Recap

0:00 – 0:20

What happens

Everyone shares where they are. We revisit key ideas from the last session to make sure they stuck. This is your chance to ask questions and surface anything that's unclear.

Why we do it

Starting a company is confusing. This ritual forces you to articulate your progress — which is the first step to understanding it. It also keeps the group connected and accountable to each other.

What's expected of you

Be present. Share honestly.

No one expects a perfect update. We're looking for genuine reflection on what you tried, what you learned, and what you're stuck on.

Progress Presentations

0:20 – 0:50

What happens

Teams present their latest progress to the group. This could be user interviews you ran, a prototype you built, data you collected, or a pivot (a change in direction) you decided on.

Why we do it

Presenting forces clarity. If you can't explain what you did and why, you haven't understood it yet. Peer feedback catches blind spots early and builds presentation skills you'll need when pitching to investors.

What's expected of you

Prepare a short update (3–5 min). Show what you did since last session, what you learned, and what's next.

Give constructive feedback to other teams. Ask real questions — the kind that help, not the kind that show off.

New Content

0:50 – 2:00

What happens

This is the teaching block. You'll learn frameworks, vocabulary, and real-world examples from startup research and practitioner experience. Every concept ties directly to what you'll build in the workshop.

Why we do it

You can't build well without understanding the problem space. This block gives you the mental models and shared language to make better decisions in the workshop and beyond.

What's expected of you

Listen actively. Ask when something's unclear — every term will be explained, and there are no stupid questions.

Take notes on anything that connects to your own startup idea. You'll use these in the workshop.

Workshop

2:00 – 3:20

What happens

This is where you build. Each module has a hands-on workshop where you apply what you just learned to your actual startup idea. Results are real deliverables: landing pages, prototypes, pitch decks.

There is no right or wrong. If you take it seriously, you can make great things. We are at your service.

Why we do it

Knowing about startups isn't the same as building one. The workshop forces you to make decisions under uncertainty — the core skill of entrepreneurship. Your deliverables become real assets for your project.



Check-Out & Beyond

3:20 – 3:30 + Home

Check-Out (in class)

Quick reflection: what did you learn today? What's still fuzzy? This closes the session and locks in your takeaways.

Optional: Case Studies & Readings

Self-paced deep dives for those who want more. Not required, but highly recommended if you're serious about building something.

Home Assignment (recommended)

Concrete group task that extends the workshop. This is where the real progress happens — in the time between sessions.

Final Presentations

At the end: you pitch in front of SpinLab. They evaluate you on team, market, and traction. Your grade is based on documented learning, not how far you got. The pitch is a milestone on the road to your first €100k in revenue.

A very good result for this course is a growing startup

Things you can control:

- 😍 Talk to 15-20 users each week
- 📈 Present you progress 3 times and ask for help
- 🎁 Offer critical feedback and questions each week
- 🔬 Be an expert in your problem space
- 🏗️ Have a pitchdeck, a landing-page and a product
- 🔄 Pivot when you hit a brick wall

Things you cannot totally control:

- 💻 Have steady growth on your landingpage
- 🖨️ Have a lead list of potential users
- 🤑 Have paying customers

Dos & Don‘ts in this entrepreneurship seminar

Thinks that are totally fine include:

- Building wild ideas
- Regrouping
- Pivoting
- Failing

Things that are not ok in here:

- Sticking with a knowingly bad idea
- Being late
- Missing class
- Staying silent when you don‘t understand

The Funding Stack

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Where the Money Comes From

Your Money

Bootstrapping

Self-funded. You keep 100% equity (ownership of the company). Growth is slower but you have total control.

Forces discipline: every euro is real, not investor money you might waste.

Viable if you can get revenue early.
Example: Basecamp, Mailchimp (early days)

Other People's Money

Angels: Wealthy individuals who invest their own money. €25k–250k. Fast decisions.

VCs (Venture Capitalists): Professional investment funds. €500k–10M+. They manage other people's money and want 10x returns.

Grants: Government money. Non-dilutive (you don't give up ownership) but slow and bureaucratic.

Accelerators: Programs that offer money + mentors + network. Usually take 5–10% equity.

Each source comes with strings attached. Know what you're trading: money for equity, speed for control, independence for support.

Equity, Dilution & How Deals Work

Equity & Dilution

You start with 100% of a company worth nothing.
Raise €500k at a €2M pre-money valuation? The investor gets 20%. You keep 80%.
Next round: another 20% gone. You're at 64%. Then 51%...

Valuation & Term Sheets

Valuation: what someone says your company is worth. Not what it is worth — it's a negotiated number.
Term sheet: the actual deal terms. Covers board seats, anti-dilution protection, and liquidation preferences (who gets paid first if the company is sold).

Convertible Notes & SAFEs

Early-stage deals often use these instead of pricing rounds. A convertible note is a loan that converts into equity later. A SAFE (Simple Agreement for Future Equity) is similar but simpler — no interest, no maturity date. Both defer the hard valuation question.

Cap Table

Short for "capitalization table." A spreadsheet showing who owns what percentage of the company. Every time you raise money or give equity to employees, the cap table changes. Keep it clean — messy cap tables scare investors away.

Source: Brad Feld & Jason Mendelson, Venture Deals (essential reading before signing any term sheet)

The Vocabulary VCs Use

TAM / SAM / SOM	TAM = Total Addressable Market (everyone who could theoretically buy). SAM = Serviceable Available Market (the slice you can realistically reach). SOM = Serviceable Obtainable Market (what you'll actually get). VCs love big TAMs.
Due Diligence	When investors check if you're telling the truth. They look at your revenue, tech, team, intellectual property, and legal structure before writing a check.
Pivot	Changing direction because reality doesn't match your hypothesis. Not failure — adaptation. Instagram started as a check-in app called Burbn.
MVP	Minimum Viable Product: the smallest thing you can build to test if anyone cares. Ship it. Learn. Repeat.
Technical Debt	Shortcuts you took to ship fast. Fine for now, expensive later. Like credit card debt for your codebase.
Go-to-Market (GTM)	How customers find you and buy. The unsexy part that makes or breaks you. A great product with no distribution is a tree falling in an empty forest.

The 8 Sectors

Sectors 1-4

Healthcare AI -- science vs speed to market

Industrial Robotics -- capital intensity + safety

Financial AI -- black-box vs regulatory demands

Creative AI -- copyright + creator backlash

Sectors 5-8

Cybersecurity -- trust + secrecy trade-offs

Climate Tech -- mission vs commercial sustainability

Legal Tech -- disrupting risk-averse buyers

Education Tech -- proving learning outcomes

Each sector has unique rivals, funding options, and dilemmas.

Sample Dilemma: A/B Test Contradiction

Your team spent months building a feature everyone loves.

The A/B test is clear: users who see it convert 15% worse.

The team insists the test must be flawed.

Options:

A) Kill the feature -- data wins, team morale takes a hit

B) Iterate based on what the data suggests

C) Trust the team's instincts, ship it anyway

There is no right answer. Welcome to entrepreneurship.

How It Works: Each Quarter

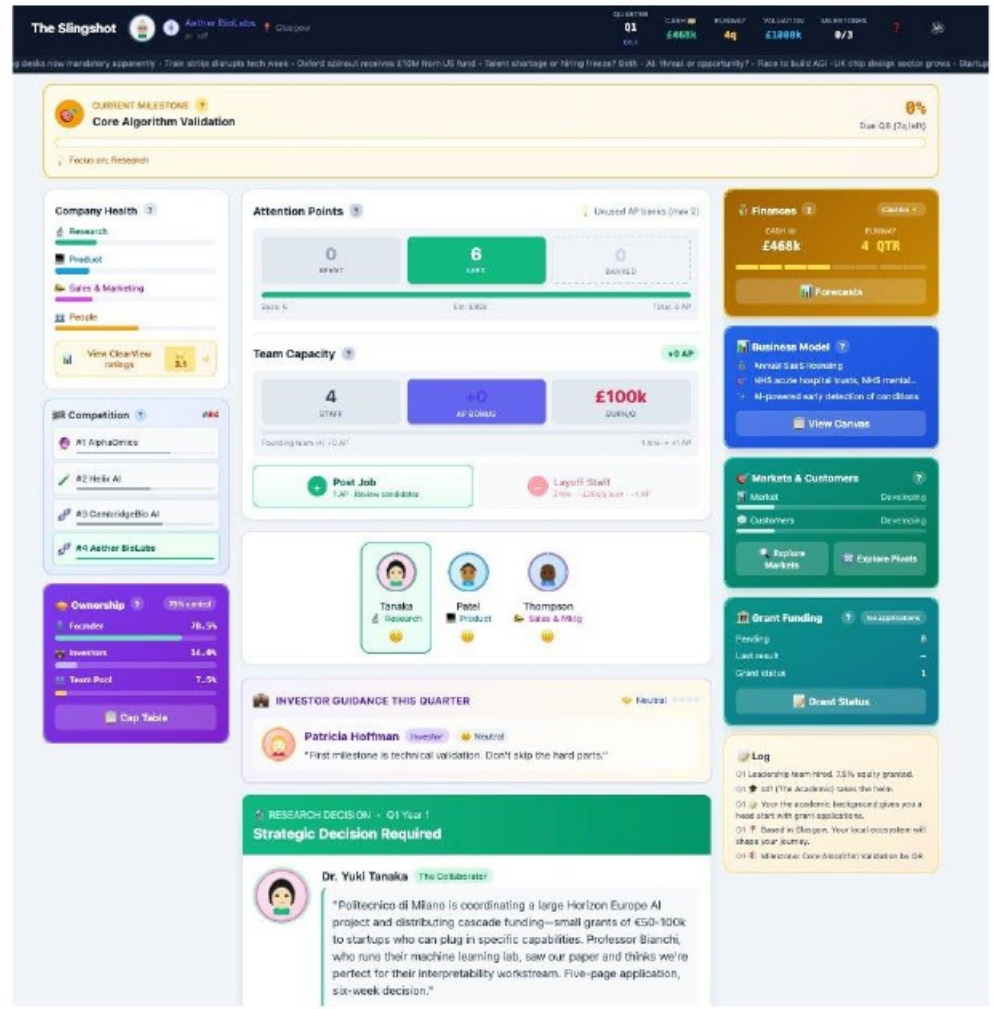
Get 6 Attention Points (AP) -- your CEO bandwidth.

Allocate AP across Research, Product, Sales, People.

You can't do everything. That's the constraint.

Bank up to 2 unused AP for next quarter.

Staff growth gives +1 to +3 bonus AP.



The Trade-offs

Publish research

+credibility, -competitive advantage

Hire fast

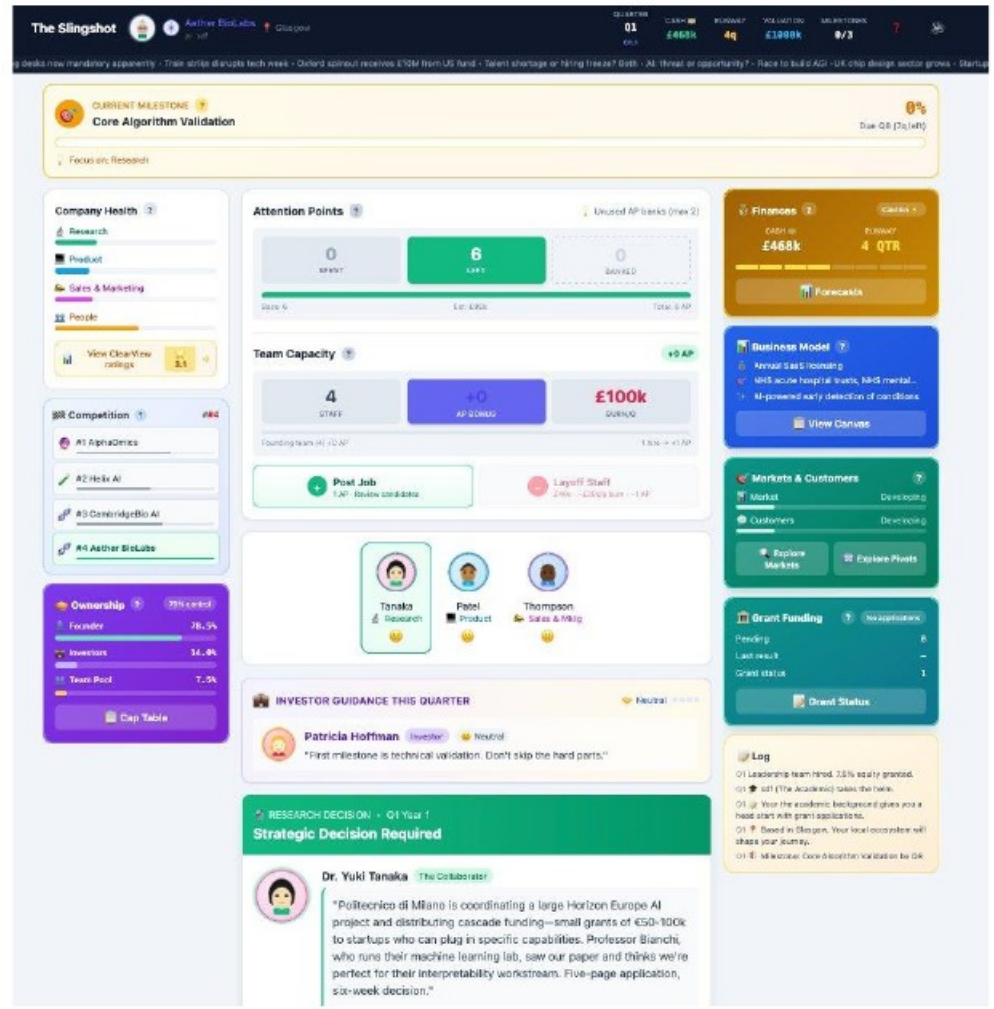
+capacity, -culture, -cash

Take VC money

+runway, -equity, -control

Ship buggy

+speed, +technical debt



The 8 Sectors

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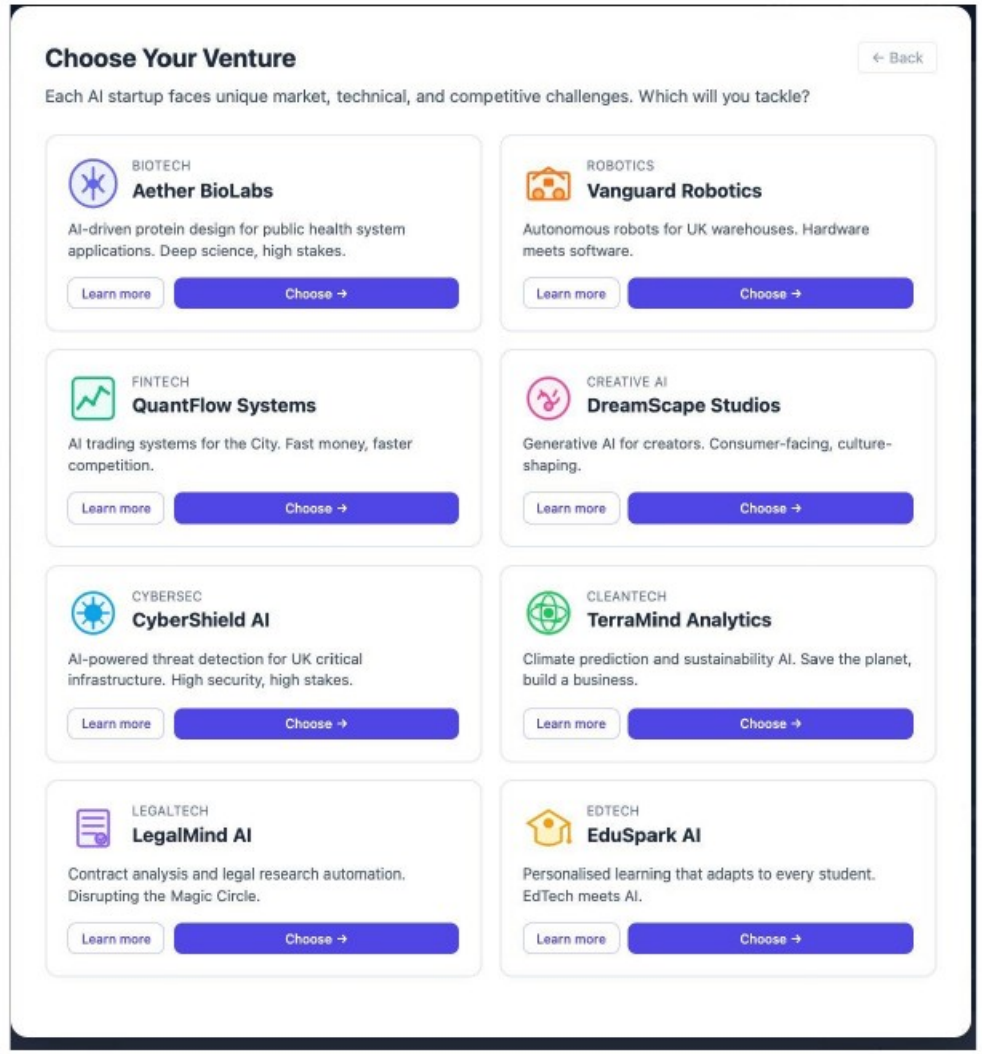
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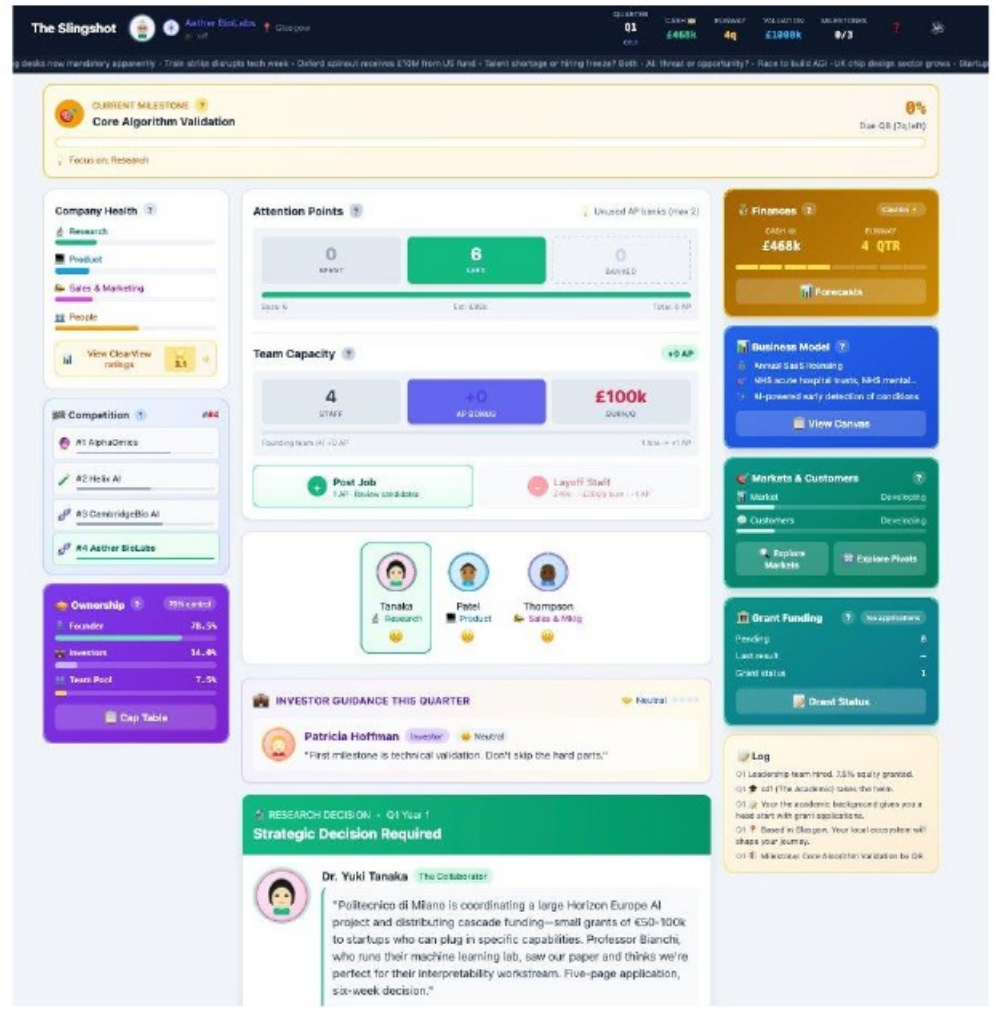
Options:

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C) Trust the team's instincts, ship it anyway

There is no right answer. Welcome to entrepreneurship.



"Good decisions improve odds but don't guarantee success. Learning to make good decisions despite uncertainty is the point."

-- The Slingshot Teaching Guide

WORKSHOP: Play The Slingshot

1. Open the game in your browser (github.com/AI-Startup-Game). No install needed.
2. Pick a sector, founder type, and location. Try Guided Play if it's your first time.
3. Play for 30 minutes. Make decisions. Watch your runway. Compete.
4. Screenshot your final results screen.
5. Debrief (15 min): What was your binding constraint? Cash, talent, market, or competition?
6. Where did you spend Attention Points early -- and what did you postpone?

This Week

Play The Slingshot at least once more. Try a different sector or founder type.

Write a one-page reflection:

-- Your strategy in one sentence

-- The biggest trade-off you faced

-- One 'oh no' moment and what you learned

-- One concept from today that explains a decision you made

Come to next session ready to share your best failure.